SUMMATIVE ASSESSMENT 2026-27

MATHEMATICS

Class : IX

1	B. $\sqrt{10}$ cm		
2	C. 9		
3	D. 6 cm		
4	D. 20		
5	B. (i) and (iii)		
6	A. 1		
7	C. Both are true. Statement 2 is the reason for statement 1.		
8	C. Both are true. Statement 2 is the reason for statement 1.		
9	Construction		
10	Let the numbers be X and Y X + Y = 33 X - Y = 15 Therefore X = 24, Y = 9 OR X + X + 15 = 33 2X = 18 X = 9 The numbers are 9 and 24		
11	(i) 5 Square units (ii) Sum of the perpendicular sides $=\sqrt{2}+\sqrt{3}$ $= 1.41 + 1.73$ $= 3.14$ Hypotenuse $=\sqrt{5}$ Square units 3.14 - 5 3.14 - 2.24 = 0.9 OR (B) (i) $\sqrt{2}$ cm (ii) $2 + 1 + \sqrt{2} + \sqrt{5}$ $= 2 + 1 + 1.41 + 2.24$ $= 6.65$ cm		

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12	Price of 4 balls = 160 (1110 - 950) Price of 1 ball = 160/4 = 40 Rupees 3 Bat + 5 x 40 = 950 Price of 3 bats =950 - 200 = 750 Price of 1 bat = 750/3 = 250 Rupees		
13	Consecutive natural numbers - X, Y XY = 1540, X+Y = 79 (X+1) (Y+1) = XY+X+Y+1 = 1540 + 79 + 1 = 1620 OR (B) 4 Consecutive natural numbers X, X+1, X+2, X+3 $X(X+3) = X^2 + 3X$ $(X+1)(X+2) = X^2 + 3X + 2$ $(X^2 + 3X + 2) - (X^2 + 3X) = 2$		
14	BD = $12 \times \frac{3}{4} = 9$ cm PD = $12 \times \frac{1}{4} = 3$ cm		
15	<table><tr><td>2</td><td>4</td></tr><tr><td>10</td><td>12</td></tr></table> 2 x 12 = 24 10 x 4 = 40 40 - 24 = 16 <table><tr><td>X</td><td>X+2</td></tr><tr><td>X+8</td><td>X+10</td></tr></table> $X(X + 10) = X^2 + 10X$ $(X+8)(X+2) = X^2 + 10X + 16$ $X^2 + 10X + 16 - (X^2 + 10X) = 16$		
16	(i) 13 = 13 x 1 $= \left(\frac{14}{2}\right)^2 - \left(\frac{12}{2}\right)^2$ $= 7^2 - 6^2$ (ii) Construction		

17	(A) (i) $l + b = 18$ $3l + 5b = 68$ (ii) $2 \times (l + b) = 36$ $l + b = 18$ .....(1) $2 \times (3l + 5b) = 136$ $3l + 5b = 68$ .....(2) Solving eq (1) & (2) $l = 11, b = 7$ (B) $\frac{X+2}{Y} = \frac{1}{2}$ ..... (1) $\frac{X}{Y-1} = \frac{1}{3}$ ..... (2) from (1) $(X + 2)\ 2 = Y \times 1$ $Y = 2X + 4$ ..... (1) from (2) $X \times 3 = 1 \times (Y - 1)$ $3X = Y - 1$ $3X = 2X + 4 - 1$ (from A) $3X = 2X + 3$ $X = 3$ $Y = 2X + 4$ $Y = 2 \times 3 + 4$ $= 10$ Fraction $= \frac{X}{Y} = \frac{3}{10}$		
18	(A) (i) PQ = 16 cm (ii) = 16 + 12 + 10 = 38 cm (iii) - When the sides are increased 4 times, the area becomes $4^2 = 16$ times the original area.		
19	(i) 44 - 16 28 cm (ii) $X + Y = 22$ $XY = 112$ $(X - 4)(Y - 4) = XY - 4(X + Y) + 16$ $= 112 - 4 \times 22 + 16$ $= 40$ Sq.cm		
20	Considering $\triangle ABC$ $\frac{AP}{PB} = \frac{AQ}{DC}$ .....(1) Considering $\triangle ACD$ $\frac{AR}{RD} = \frac{AQ}{QC}$ .....(2)		

	From (1) & (2) $\frac{AR}{RD} = \frac{AQ}{QC} = \frac{AR}{RD}$ $\frac{AP}{PB} = \frac{AR}{RD}$ OR B. AP = $\frac{1}{2}$ AC (P Midpoint) QR = $\frac{1}{2}$ AC AP = QR .....(1) PQ = $\frac{1}{2}$ AB AR = $\frac{1}{2}$ AB PQ = AR .....(2) < A = 90^0 PQ is parallel to AB QR parallel to AC < P = < Q = <R = 90^0 The opposite sides are equal and parallel and all the angles are 90^0. Therefore, ARPQ is a square. (ii) PQ = X OR = Y ആയാൽ Area of △ ABC = $\frac{1}{2}$ x 2X x 2Y = 2XY Area of rectangle = XY Therefore, the area of the rectangle is $\frac{1}{2}$ of the area of △ABC		
21	Construction		
22	(i) $\sqrt{3}$ cm (ii) = 2 + 1 +$\sqrt{3}$ = 3 + 1.73 = 4.73 cm (iii) Perimeter of △ ABC = 2 + 2 + 2 = 6 cm Therefore, 6 - 4.73 = 1.27 cm OR B. (i) $\sqrt{2}$ (ii) $\sqrt{2}$ + 1 +$\sqrt{3}$ = 1.41 + 1 + 1.73 = 4.14 (iii) 10, 1$\sqrt{11}$		

23	(i) Larger number = X, Smaller number = Y (X+1) (Y-1) = 390 XY - X + Y - 1 = 390.....(1) (X - 1) (Y + 1) = 408 XY + X - Y - 1 = 408.....(2) (1) + (2) => 2XY - 2 = 798 2XY = 800 XY = 400 (ii) In eq (2) 400 + X - Y - 1 = 408 399 + X - Y = 408 X - Y = 408 - 399 = 9		
24	(A) PQRB is a parallelogram. PQ is parallel to BC PQ = $\frac{1}{2}$ BC .....(1) BR = $\frac{1}{2}$ BC .....(2) From (1) & (2) PQ = BR Since one pair of opposite sides are equal and parallel, PQRB is a parallelogram. The diagonals of a parallelogram bisect each other. PS = SR That is, BQ bisects PR. B. (i) 6 : 4 = 3 : 2 (ii) BP = 15 x $\frac{3}{5}$ = 9 cm PC = 15 x $\frac{2}{5}$ = 6 cm (iii) 3 : 2		
25	Price of a bag = X Price of an umbrella = Y 7X + 3Y = 3690 .....(1) 3X + 7Y = 3010 .....(2) (1) + (2) 10X + 10Y = 6700 X + Y = 670		

(1) - (2) => 4X - 4Y = 680

X - Y = 170

X + Y = 670 +

X - Y = 170

2X + 0 = 840

X = 420

Y = 670 - 420

= 250

26 A. Let the numbers be X, Y

(X+2) (Y+2) = 1053

XY + 2(X+Y) + 4 = 1053

(X-2) (Y-2) = 805

XY - 2 (X + Y) + 4 = 805

XY + 2 (X + Y) + 4 = 1053 +

XY - 2 (X + Y) + 4 = 805

2XY + 0 + 8 = 1858

2XY = 1850

XY = 1850/2 = 925

XY + 2(X+Y) + 4 = 1053

925 + 2 (X+Y) + 4 = 1053

925 + 2 (X+Y) = 1053

2 (X+Y) = 1053 - 929

2 (X+Y) = 124

X + Y = 62

OR

B.



(i) (X-1) (Y+1) = XY + 16

~~XY~~ + X - Y - 1 = ~~XY~~ + 16

X - Y = 17

(X+1) (Y-1) = XY - X+Y - 1

= XY - (X-Y) -1

= XY - 17 - 1

= XY - 18

Therefore, the area decreases by 18 square meters.

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